Fiscal Year:	FY 2018	**Task Last Updated:**	FY 05/23/2018
PI Name:	Law, Chung K. Ph.D.		
Project Title:	Structure and Response of Spherical Diffusion Flames		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	Chung Law's email is cklaw@princeton.edu		
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	11/25/2014	**End Date:**	11/24/2019
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 11/24/2019 per NSSC info (Ed., 11/25/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Tse, Stephen Ph.D. (Rutgers University)		
Grant/Contract No.:	NNX15AC79A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	NOTE: Continuation of Structure and Response of Spherical Diffusion Flames--NNX09AR03A The effort was divided into experimental and theoretical elements. The experimental effort consisted of using a porous spherical burner to issue fuel mixtures into an oxidizing environment. Different mixtures of fuel species permitted investigation of the phenomena of interest. The experiment effort was conducted primarily in the drop towers at the Glenn Research Center (GRC) and the NASA reduced-gravity aircraft. Space flight experiment was fully defined and approved during the program period. The theoretical effort consisted of the development of computational codes simulating the combustion phenomena of interest, and interpreting the calculated results on the basis of fundamental flame physics. The purpose of this flight definition task is to gain fundamental understanding of the structure and dynamics of diffusion flames in a simple, well-defined flow field. Using a spherical geometry this work considers: 1) gas-phase transient processes, including fuel vapor accumulation inside the flame and far-field diffusion, and their		

Task Description:	influence on flame movement and the achievement of steady flame behavior; 2) extinction boundaries of steady or near-steady flames and the identification of the associated chemistry at the global, reduced, and detailed levels; 3) the characteristics of partially-premixed flames; 4) flamefront pulsation and cellular instability especially near the extinction limits; 5) the dynamics of rotating flames; and 6) the behavior and structure of quasi-spherical flames in partial gravity environments. These studies are intended to significantly enhance the understanding of diffusion flames, especially providing capability for predicting flame behavior under varied conditions and configurations. As such, this work should contribute to engineering approaches to practical combustion systems, achieving increased energy conversion efficiencies and reductions in unwanted combustion emissions in both terrestrial and extraterrestrial environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Enhanced understanding of flame-flow vortical interaction could lead to improved burner design and operation in terms of energy efficiency and pollutant emissions.
Task Progress:	Initiation and propagation of laminar premixed cool flames: Initiation and propagation of premixed cool flames for DME/O2/N2 mixtures are computationally studied in the 1-D planar configuration as well as the counterflow configuration. The results demonstrate that a residence time window exists for the occurrence of cool flames; external to this window a cool flame either cannot be initiated or will transition to a hot flame. Parametric study shows that the cool flame regime is sensitive to the amount of dilution in the flow, and that the cool flame becomes more intense at richer conditions. Global parameters such as the reference cool flame speed and cool flame temperature have been studied in a symmetric counterflow for various strain rates and equivalence ratios, with the controlling reactions identified. The results show that the premixed cool flame speeds are relatively low, around 6-7 cm/s, and are much less sensitive in terms of equivalence ratio variations. Experimental investigations have also been carried out in a counterflow, with a steady cool flame observed with a CCD camera with high sensitivity of the UV band. Such observations qualitatively demonstrate stronger cool flame chemiluminescence and insensitive change in the flame position with increasing equivalence ratios, which correspond well with the detailed numerical simulation. This work is reported in: “Initiation and propagation of laminar premixed cool flames,” Peng Zhao, Wenkai Liang, Sili Deng, and Chung K. Law, Fuel 166 (2016) 477-487. Aspects of Colliding Spherical Flames: The morphology, corner dynamics, and flame- generated vorticity in colliding spherical flames are investigated. Schlieren images of the flame surface show that the morphology of the colliding process can be divided into three stages, which is further supported by simulations. Using experimental and simulation results, a comprehensive understanding of the collision dynamics is acquired, particularly the corner dynamics and the vortices generated near the corner. Regarding the former, we have identified strongly interacting factors, namely preferential diffusion and kine- matic restoration. Regarding the latter, we have demonstrated that the vorticity is generated by the baroclinic torque across the flame as a result of hydrodynamic pressure redistribution and density jump. The magnitude of the vorticity is estimated under quasi-steadiness assumption. By comparing the scales of the kinetic energy of the flame-generated vortices with those of the Kolmogorov eddies, it is suggested that vorticity at the flame corner could persist over dissipation and accumulate to a substantial extent to affect the turbulence spectrum as well as the local/global burning rate. This work is reported in: “On Colliding Spherical Flames: Morphology, Corner Dynamics, and Flame-generated Vorticity,” Sheng Yang, Delin Zhu, and Chung K. Law, Combustion and Flame 167 (2016) 444-451. Extended Flammability Limits of n-Heptane/Air Mixtures with Cool Flames: A numerical and theoretical investigation has been conducted on the flammability limits of n-heptane/air mixtures by considering the role of the cool flame chemistry. It is found that the existence of the cool flame can substantially extend the lean and rich limits of the conventional high-temperature flame, with the extended limits being insensitive to pressure but sensitive to the extent of inert dilution. Different levels of kinetic criteria based on the controlling branching and termination reactions at low temperatures are derived and shown to agree well with the numerically simulated results at normal pressure. Furthermore, the critical temperature of cool flame flammability limits at normal pressure is found to be around 630 K and increases with pressure due to the shift of the NTC regime to higher temperatures.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Liang W, Law CK. "Extended flammability limits of n-heptane/air mixtures with cool flames." Combustion and Flame. 2017 Nov;185:75-81. https://doi.org/10.1016/j.combustflame.2017.06.015 , Nov-2017
Articles in Peer-reviewed Journals	Liang W, Law CK. "Theory of first-stage ignition delay in hydrocarbon NTC chemistry." Combustion and Flame. 2018 Feb;188:162-9. https://doi.org/10.1016/j.combustflame.2017.10.003 , Feb-2018